Data Science Virtual Internship : Customer Booking & Lounge Analysis

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Project Overview

The primary objective of this analysis was to build a machine learning model capable of predicting the likelihood of a customer completing a holiday booking. By identifying key drivers of conversion, British Airways can optimize marketing strategies and improve sales performance.

Lounge Eligibility Analysis

BRITISH AIRWAYS

Lounge Eligibility Lookup

Grouping	Example Destinations	Tier 1 %	Tier 2 %	Tier 3 %	Notes
Long-haul AM	Singapore, Hong Kong, Tokyo	1.5%	20.5%	78%	High demand for morning international flights
Long-haul PM	New York, Los Angeles, San Fransisco	1.5%	20.4%	78%	Consistent loyalty mix for overnight routes
Short-haul AM	Paris, Berlin, Amsterdam	1.6%	20.4%	78%	Morning business regional commuters
Short-haul PM	Madrid, Rome, Barcelona	1.6%	20.4%	78%	Steady leisure and regional travel patterns
Europe	Paris, Frankfurt, Brussels	1.6%	20.4%	78%	Largest volume for regional short-haul
North America	New York, Chicago, Miami	1.5%	20.5%	78%	Premium long-haul loyalty distribution

Data Pipeline & Quality Assurance

Data Volume : The analysis initiated with a raw dataset of **50,000** customer transactions.

Data Cleaning : Identified and removed **719 duplicate records**, resulting in a clean dataset of **49,281** rows to ensure model integrity and prevent bias.

Feature Engineering: Performed mapping on the flight_day variable and categorical encoding to improve the algorithm's computational performance.

Validation Strategy: Employed an **80:20 data split** (Training: 39,424 | Testing: 9,857) to evaluate the model's accuracy on unseen data.

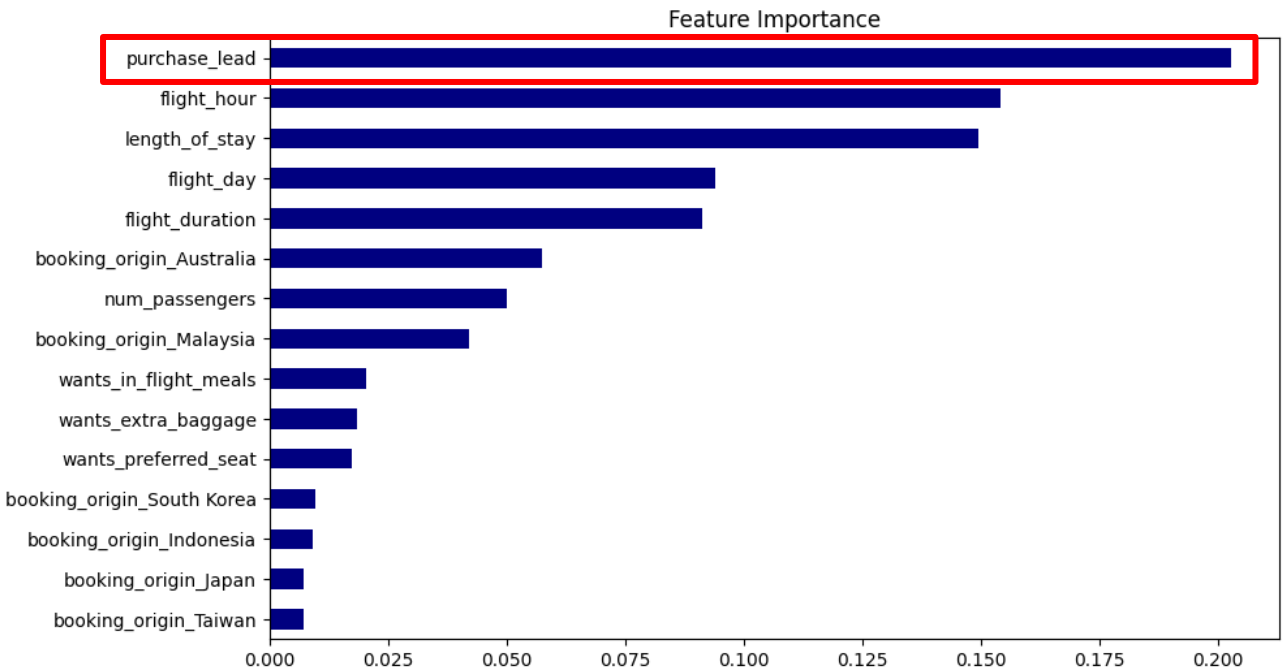
Predictive Modeling Overview

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 14 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   num_passengers         50000 non-null  int64
 1   sales_channel          50000 non-null  object
 2   trip_type              50000 non-null  object
 3   purchase_lead          50000 non-null  int64
 4   length_of_stay         50000 non-null  int64
 5   flight_hour            50000 non-null  int64
 6   flight_day             50000 non-null  object
 7   route                  50000 non-null  object
 8   booking_origin         50000 non-null  object
 9   wants_extra_baggage    50000 non-null  int64
 10  wants_preferred_seat   50000 non-null  int64
 11  wants_in_flight_meals  50000 non-null  int64
 12  flight_duration        50000 non-null  float64
 13  booking_complete       50000 non-null  int64
dtypes: float64(1), int64(8), object(5)
memory usage: 5.3+ MB
```

```
# Recheck
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49281 entries, 0 to 49280
Data columns (total 13 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   num_passengers         49281 non-null  int64
 1   sales_channel          49281 non-null  object
 2   trip_type              49281 non-null  object
 3   purchase_lead          49281 non-null  int64
 4   length_of_stay         49281 non-null  int64
 5   flight_hour            49281 non-null  int64
 6   flight_day             49281 non-null  object
 7   booking_origin         49281 non-null  object
 8   wants_extra_baggage    49281 non-null  int64
 9   wants_preferred_seat   49281 non-null  int64
 10  wants_in_flight_meals  49281 non-null  int64
 11  flight_duration        49281 non-null  float64
 12  booking_complete       49281 non-null  int64
dtypes: float64(1), int64(8), object(4)
memory usage: 4.9+ MB
```

- There are 50000 raw data
- Remove 719 duplicates data
- Using 80:20 splitting data

Feature Importance



Based on the **Random Forest** model, three primary factors were found to significantly influence a customer's decision:

- **purchase_lead**: The most dominant variable, indicating that the time between booking and departure is the strongest predictor of conversion.
- **flight_hour**: The hour of departure has a significant impact on booking completion.
- **length_of_stay**: The duration of the trip is a critical consideration for customers before making a final payment.

Model Performance & Recommendations

```
# Classification Report
print(classification_report(y_test, model.predict(X_test)))
```

	precision	recall	f1-score	support
0	0.86	0.98	0.92	8378
1	0.48	0.08	0.14	1479
accuracy			0.85	9857
macro avg	0.67	0.53	0.53	9857
weighted avg	0.80	0.85	0.80	9857

Model Performance Evaluation

- **Overall Accuracy** : The model achieved a solid overall accuracy of **85%**.
- **Precision (Class 1)** : For customers who completed a booking, the model achieved a **Precision of 0.48**, meaning nearly half of the "will buy" predictions were accurate.
- **Class 0 Performance** : The model is exceptionally strong at identifying customers who will not complete a booking, with an **F1-score of 0.92**.
- **Analysis of Recall** : A **Recall of 0.08** highlights the challenge of imbalanced data, suggesting that customers who complete bookings have highly specific patterns that require additional behavioral data (such as search history or loyalty status) for better detection.

Conclusions

Data-Drive Insights	The analysis successfully identified that **Purchase Lead**, **Flight Hour**, and **Length of Stay** are the primary drivers of booking completions.
Predictive Accuracy	The Random Forest model achieved a robust **85% accuracy**, providing a reliable foundation for identifying high-potential customers.
Operational Sustainability	**Task 1** analysis confirms that **Tier 1 & 2** lounge eligibility is sustainable, as these premium segments represent a manageable fraction of total passengers.
Model Integration	There is a clear link between customer loyalty (Task 1) and booking behavior (Task 2), suggesting that integrated data is key to British Airways' growth.

Strategic Recommendations

Targeted Outreach	Focus marketing efforts on customers with a long purchase_lead, as they demonstrate a higher intent to finalize bookings.
Booking Incentives	Provide specific incentives or reminders during peak flight_hour windows to encourage transaction completion.
Model Enhancement	Incorporate customer profile features (such as Loyalty Tier Status from Task 1) to increase the model's sensitivity in detecting potential buyers.